

DATA CLEANING & REPORTING AUTOMATION

Project Report

Project Title	Data Cleaning & Reporting Automation
Domain	Data Analytics
Tools	Python / Excel / Power BI
Main Objective	Automate data cleaning and generate clear reports and visual summaries

1. Introduction

Data Cleaning & Reporting Automation is a data analytics project that focuses on preparing raw data for analysis and producing useful reports automatically. Real-world datasets may contain missing values, duplicate records, inconsistent formats, and incorrect or incomplete entries. Cleaning these issues improves data quality and makes the final analysis more reliable.

2. Problem Statement

Manual data cleaning and report preparation can take significant time and may introduce human errors. The project aims to create a simple workflow that identifies and handles common data-quality problems and then produces an automated report with summary information and visual insights.

3. Objectives

- Identify missing values and handle them appropriately.
- Detect and remove duplicate records.
- Find and correct inconsistent data formats and values.
- Prepare a clean dataset for analysis.
- Generate summary statistics and visual reports.
- Reduce repetitive manual work through automation.

4. Tools and Technologies

Tool	Purpose
Python	Data cleaning, transformation, analysis, and automation
Pandas	Reading, cleaning, and processing tabular data
Excel	Data inspection, formulas, filtering, and reporting
Power BI	Interactive dashboards and visual summaries

5. Project Workflow

- Collect or import the raw dataset.
- Inspect columns, data types, missing values, and duplicate records.
- Clean missing values and remove duplicate entries.
- Standardize inconsistent values, spelling, and formats.
- Validate the cleaned dataset.
- Generate summary statistics and visualizations.
- Export the cleaned data and final report.

6. Data Cleaning Process

Missing Values: Missing entries are identified and handled using suitable methods such as replacement with an appropriate value or removal when necessary.

Duplicate Records: Repeated rows are detected and removed to avoid counting the same information more than once.

Inconsistent Data: Values such as category names, text formats, dates, and numeric fields are standardized so that they can be analyzed consistently.

7. Reporting and Visualization

After cleaning, the dataset can be summarized using tables, charts, KPIs, and dashboards. Visual reports help users understand trends, distributions, and important patterns without manually checking every row of the dataset.

8. Expected Output

- Cleaned and validated dataset.
- Summary statistics.
- Automated report containing important findings.
- Charts or dashboard showing key patterns.
- Reusable workflow that reduces manual reporting effort.

9. Benefits

- Improves data quality and consistency.
- Saves time during repetitive data preparation.
- Reduces errors caused by manual processing.
- Makes reporting faster and more organized.
- Supports better understanding of business or project data.

10. Conclusion

This project demonstrates how data cleaning and reporting tasks can be organized into an automated workflow. By handling missing values, duplicates, and inconsistent data before analysis, the quality of the final report is improved. The project also provides practical experience with data preprocessing, automation, and visualization.

11. GitHub Submission Structure

The following files can be uploaded to the GitHub repository along with this report:

- README.md – project overview, objectives, tools, workflow, and results.
- Raw dataset – original sample data, if permitted by the task.
- Cleaned dataset – processed data after cleaning.
- Python/Excel/Power BI files – implementation files.
- Report PDF – this project report.
- Screenshots – optional evidence of cleaning steps and dashboard/report output.